

Summer School 2025
Astronomy & Astrophysics



Project Report

Prepared By

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Projects Name

ISS Python Tracking

Submitted To

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Designation: Program Supervisor
Institution: India Space Academy

Importing Libraries

```
In [65]: # Import necessary Libraries

import requests
import numpy as np
import urllib.request
import json
import geocoder
import time
import geopy
import matplotlib.pyplot as plt
```

Fetching ISS (Two line element) TLE data

```
In [71]: response = urllib.request.urlopen('http://api.open-notify.org/iss-now.json')
result = json.loads(response.read())
print(result)

{'timestamp': 1753380919, 'iss_position': {'longitude': '143.4226', 'latitude': '-47.2992'}, 'message': 'success'}
```

```
In [72]: from datetime import timedelta, datetime
from geopy.geocoders import Nominatim
import pytz
ist = pytz.timezone('Asia/Kolkata')

# Initialize Nominatim API
geolocator = Nominatim(user_agent="my_geopy_app")
name = str(input("Enter a location name: "))
location = geolocator.geocode(name)
print("\nLocation Coordinates:")
print(f"Latitude: {location.latitude}")
print(f"Longitude: {location.longitude}\n")

latitude, longitude = location.latitude, location.longitude
from skyfield.api import EarthSatellite, Topos, load

# TLE Lines
line1 = '1 25544U 98067A   25204.16864096   .00010724   00000+0  19544-3 0  9992'
line2 = '2 25544   51.6344 128.7280 0002115 106.2807 253.8414 15.50025731520740'
satellite = EarthSatellite(line1, line2, 'ISS (ZARYA)')

# Load timescale and observer Location
ts = load.timescale()
observer = Topos(latitude_degrees=latitude, longitude_degrees=longitude)
t0 = ts.now()
t1 = ts.utc(t0.utc_datetime() + timedelta(days=0.5)) # Next 12 hours

# Find events (rise, culmination, set)
times, events = satellite.find_events(observer, t0, t1, altitude_degrees=0.0)

event_names = ['rise', 'culminate', 'set']
print("Next ISS Passes:\n")
group = []
for ti, event in zip(times, events):
    utc_dt = ti.utc_datetime().replace(tzinfo=pytz.utc)
    ist_dt = utc_dt.astimezone(ist)
    group.append(f"{event_names[event].capitalize()}: {ist_dt.strftime('%Y-%m-%d %H:%M:%S IST')}")
    if event == 2: # 'set' event
        print("\n".join(group))
        print("-" * 40)
        group = []
```

Location Coordinates:
Latitude: 21.2380912
Longitude: 81.6336993

Next ISS Passes:

Rise: 2025-07-25 11:06:35 IST
Culminate: 2025-07-25 11:06:56 IST
Set: 2025-07-25 11:07:17 IST

```
In [20]: from mpl_toolkits.basemap import Basemap
```

```
In [73]: plt.figure(figsize=(12, 6))

m = Basemap(projection='cyl', llcrnrlat=-90, urcnrlat=90,
            llcrnrlon=-180, urcnrlon=180, resolution='c')
m.drawcoastlines()
m.fillcontinents(color='grey')
m.drawparallels(np.arange(-90., 91., 30.))
m.drawmeridians(np.arange(-180., 181., 60.))
m.drawmapboundary(fill_color='lightblue')

# ISS ground track
m.plot(lons, lats, color='red', linewidth=1, label='ISS Ground Track')

# Mark current position (first point)
m.scatter(lons[0], lats[0], color='blue', s=80, label='Current Position', zorder=5)
```

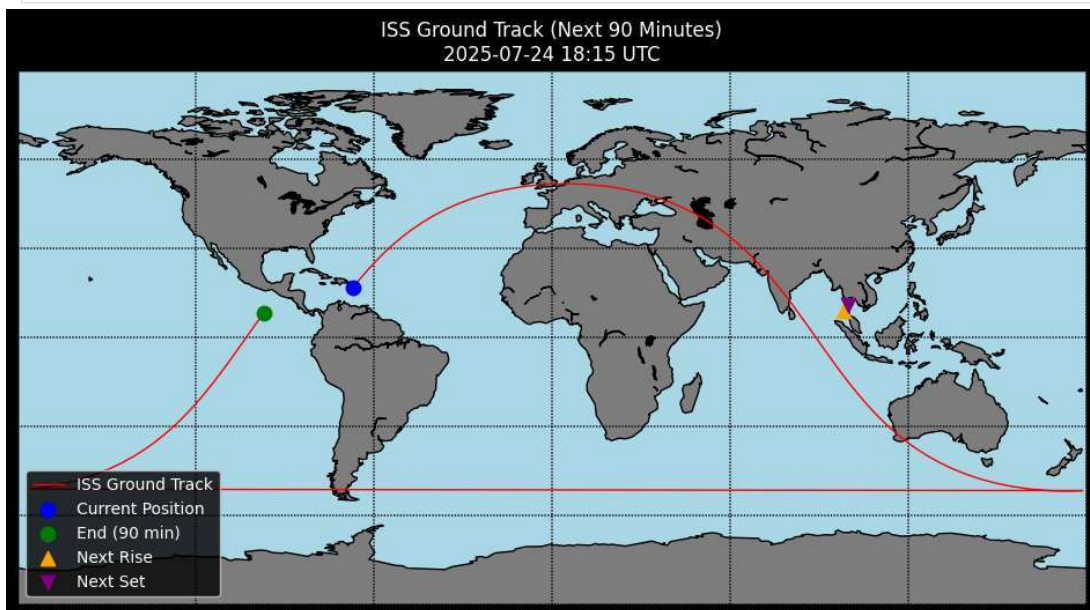
```

# Mark end position (last point)
m.scatter(lons[-1], lats[-1], color='green', s=80, label='End (90 min)', zorder=5)

# Mark next rise and set points if available
if 'times' in locals() and 'events' in locals():
    # Find next rise and set events
    rise_idx = None
    set_idx = None
    for idx, event in enumerate(events):
        if event == 0 and rise_idx is None: # rise
            rise_idx = idx
        if event == 2 and set_idx is None: # set
            set_idx = idx
    if rise_idx is not None:
        rise_time = times[rise_idx]
        rise_geocentric = satellite.at(rise_time)
        rise_subpoint = rise_geocentric.subpoint()
        m.scatter(rise_subpoint.longitude.degrees, rise_subpoint.latitude.degrees,
                  color='orange', s=80, marker='^', label='Next Rise', zorder=5)
    if set_idx is not None:
        set_time = times[set_idx]
        set_geocentric = satellite.at(set_time)
        set_subpoint = set_geocentric.subpoint()
        m.scatter(set_subpoint.longitude.degrees, set_subpoint.latitude.degrees,
                  color='purple', s=80, marker='v', label='Next Set', zorder=5)

# Title with current date/time
from datetime import datetime
plt.title(f'ISS Ground Track (Next 90 Minutes)\n{datetime.utcnow().strftime('%Y-%m-%d %H:%M UTC')}')
plt.legend(loc='lower left')
plt.show()

```



```

In [70]: import pandas as pd

# Example CSV: Locations.csv with columns: name,latitude,longitude
# name,latitude,longitude
#Delhi 28.7 77.1
#Raipur 21.25 81.62
#Ahmedabad 23.02 72.57
#Bhopal 23.25 77.41
#Nainital 29.39 79.45

df = pd.read_csv('locations.csv')
results = []

for idx, row in df.iterrows():
    observer = Topos(latitude_degrees=row['latitude'], longitude_degrees=row['longitude'])
    t0 = ts.now()
    t1 = ts.utc(t0.utc_datetime() + timedelta(days=1))
    times, events = satellite.find_events(observer, t0, t1, altitude_degrees=0.0)
    event_names = ['rise', 'culminate', 'set']
    group = {}
    for ti, event in zip(times, events):
        utc_dt = ti.utc_datetime().replace(tzinfo=pytz.utc)
        ist_dt = utc_dt.astimezone(ist)
        group[event_names[event]] = ist_dt.strftime('%Y-%m-%d %H:%M:%S IST')
        if event == 2: # set
            results.append({
                'Location': row['name'],
                'Rise Time': group.get('rise', ''),
                'Culmination': group.get('culminate', ''),
                'Set Time': group.get('set', ''),
                'Duration (min)': (
                    pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))
                ).total_seconds() / 60 if group.get('rise', '') and group.get('set', '') else ''
            })
    group = {}

```

```

results_df = pd.DataFrame(results)
print(results_df)
results_df.to_csv('iss_passes_results.csv', index=False)

```

	Location	Rise Time	Culmination
0	Delhi	2025-07-25 12:39:21 IST	2025-07-25 12:44:05 IST
1	Delhi	2025-07-25 14:15:23 IST	2025-07-25 14:20:36 IST
2	Delhi	2025-07-25 15:55:23 IST	2025-07-25 15:58:13 IST
3	Delhi	2025-07-25 20:49:29 IST	2025-07-25 20:54:07 IST
4	Delhi	2025-07-25 22:25:41 IST	2025-07-25 22:31:02 IST
5	Raipur	2025-07-25 11:06:41 IST	2025-07-25 11:06:57 IST
6	Raipur	2025-07-25 12:37:46 IST	2025-07-25 12:43:09 IST
7	Raipur	2025-07-25 14:16:08 IST	2025-07-25 14:20:05 IST
8	Raipur	2025-07-25 20:52:21 IST	2025-07-25 20:56:19 IST
9	Raipur	2025-07-25 22:27:53 IST	2025-07-25 22:33:16 IST
10	Ahmedabad	2025-07-25 00:54:04 IST	2025-07-25 00:56:05 IST
11	Ahmedabad	2025-07-25 12:37:28 IST	2025-07-25 12:42:12 IST
12	Ahmedabad	2025-07-25 14:13:38 IST	2025-07-25 14:18:44 IST
13	Ahmedabad	2025-07-25 20:51:29 IST	2025-07-25 20:54:07 IST
14	Ahmedabad	2025-07-25 22:26:06 IST	2025-07-25 22:31:27 IST
15	Bhopal	2025-07-25 12:37:49 IST	2025-07-25 12:42:57 IST
16	Bhopal	2025-07-25 14:14:54 IST	2025-07-25 14:19:39 IST
17	Bhopal	2025-07-25 20:51:24 IST	2025-07-25 20:55:06 IST
18	Bhopal	2025-07-25 22:26:46 IST	2025-07-25 22:32:11 IST
19	Nainital	2025-07-25 12:39:39 IST	2025-07-25 12:44:34 IST
20	Nainital	2025-07-25 14:16:00 IST	2025-07-25 14:21:09 IST
21	Nainital	2025-07-25 15:56:07 IST	2025-07-25 15:58:51 IST
22	Nainital	2025-07-25 19:14:40 IST	2025-07-25 19:16:32 IST
23	Nainital	2025-07-25 20:49:33 IST	2025-07-25 20:54:28 IST
24	Nainital	2025-07-25 22:26:01 IST	2025-07-25 22:31:14 IST

	Set Time	Duration (min)
0	2025-07-25 12:48:50 IST	9.483333
1	2025-07-25 14:25:50 IST	10.450000
2	2025-07-25 16:01:04 IST	5.683333
3	2025-07-25 20:58:45 IST	9.266667
4	2025-07-25 22:36:20 IST	10.650000
5	2025-07-25 11:07:13 IST	0.533333
6	2025-07-25 12:48:32 IST	10.766667
7	2025-07-25 14:24:04 IST	7.933333
8	2025-07-25 21:00:16 IST	7.916667
9	2025-07-25 22:38:38 IST	10.750000
10	2025-07-25 00:58:06 IST	4.033333
11	2025-07-25 12:46:56 IST	9.466667
12	2025-07-25 14:23:51 IST	10.216667
13	2025-07-25 20:56:45 IST	5.266667
14	2025-07-25 22:36:47 IST	10.683333
15	2025-07-25 12:48:05 IST	10.266667
16	2025-07-25 14:24:26 IST	9.533333
17	2025-07-25 20:58:48 IST	7.400000
18	2025-07-25 22:37:35 IST	10.816667
19	2025-07-25 12:49:29 IST	9.833333
20	2025-07-25 14:26:19 IST	10.316667
21	2025-07-25 16:01:35 IST	5.466667
22	2025-07-25 19:18:24 IST	3.733333
23	2025-07-25 20:59:21 IST	9.800000
24	2025-07-25 22:36:25 IST	10.400000

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 12:48:50 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 12:39:21 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 14:25:50 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 14:15:23 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 16:01:04 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 20:58:45 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 20:49:29 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 22:36:20 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 22:25:41 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

pd.to_datetime(group.get('set', '')) - pd.to_datetime(group.get('rise', ''))

C:\Users\asus\AppData\Local\Temp\ipykernel_11032\1760228499.py:33: FutureWarning: Parsed string "2025-07-25 11:07:13 IST" in cluded an un-recognized timezone "IST". Dropping unrecognized timezones is deprecated; in a future version this will raise.